W12 HARD

SCENARIO PRACTICE

IoT Data Analytics + ML + Big Data + IoT Security

WHAT THIS PDF TRAINS

- Long 8-mark analytics scenarios
- DDPP identification from clues
- Scaling vs Volatility database problems
- Local vs Remote ML processing
- MBOS application-domain scenarios
- Big Data 3V and IoT security cases
- Hard integrated design + traps + rapid questions

Priority First

DDPP -> SV -> Local/Remote -> MBOS -> 3V -> Security 8

Answer formula: Identify -> Definition -> Scenario clue -> 5-7 points -> Conclusion

Scenario 1: Factory Dashboard

Long 8-mark pattern - identify, justify, explain

SCENARIO

A factory dashboard continuously displays current temperature, pressure, vibration and machine status from hundreds of IoT sensors. The manager mainly wants to know what is happening right now.

QUESTION

Which type of IoT data analytics is being used?

Best answer: Descriptive Analytics

- Descriptive analytics explains the current situation.
- It answers the question: 'What is happening?'
- It summarizes or presents available data.
- In IoT, current sensor readings and dashboards are natural descriptive examples.

MEMORY

Descriptive = Describe the current condition.

Scenario 2: Machine Failure Investigation

Long 8-mark pattern - identify, justify, explain

SCENARIO

A pump failed during operation. Engineers analyze its temperature, pressure and vibration logs to understand the reason for the failure.

QUESTION

Which analytics type fits this task?

Best answer: Diagnostic Analytics

- Diagnostic analytics focuses on finding the reason behind an event.
- It answers: 'Why did it happen?'
- It is used for fault or root-cause investigation.
- It comes after descriptive analysis in the analytics sequence.

MEMORY

Diagnostic = Doctor finds the cause.

Scenario 3: Predictive Maintenance

Long 8-mark pattern - identify, justify, explain

SCENARIO

A machine is still operating, but an analytics system studies previous and current sensor readings to estimate whether the machine is likely to fail soon.

QUESTION

Which analytics type is being used?

Best answer: Predictive Analytics

- Predictive analytics estimates a future outcome.
- It answers: 'What is likely to happen?'
- It can use sensor history and patterns.
- A common IoT use is predicting possible failure before it happens.

MEMORY

Predictive = Future.

Scenario 4: Recommended Corrective Action

Long 8-mark pattern - identify, justify, explain

SCENARIO

After predicting a likely machine failure, the system recommends reducing the machine load and scheduling maintenance immediately.

QUESTION

Which analytics type is represented?

Best answer: Prescriptive Analytics

- Prescriptive analytics recommends what action should be taken.
- It answers: 'What should I do about it?'
- It goes beyond prediction and supports decisions.
- It is shown as the most advanced stage in the four-type analytics flow.

MEMORY

Prescriptive = Action.

Scenario 5: Full DDPP Chain

Long 8-mark pattern - identify, justify, explain

SCENARIO

An industrial IoT platform shows current condition, investigates yesterday's fault, estimates tomorrow's failure probability, and then recommends maintenance action.

QUESTION

Match all four tasks with the four analytics types.

Best answer: Descriptive -> Diagnostic -> Predictive -> Prescriptive

- Current condition -> Descriptive.
- Reason for past fault -> Diagnostic.
- Future failure estimate -> Predictive.
- Recommended action -> Prescriptive.
- The sequence moves from understanding the present/past toward future decision-making.

MEMORY

DDPP = Descriptive - Diagnostic - Predictive - Prescriptive | What - Why - Future - Action.

Scenario 6: Value and Complexity

Long 8-mark pattern - identify, justify, explain

SCENARIO

A company starts with a simple current-status dashboard, then adds root-cause analysis, future prediction, and finally decision recommendations.

QUESTION

How do value and complexity change according to the W12 slide?

Best answer: Both increase toward Prescriptive Analytics

- The graph places Descriptive at the lowest value/complexity end.
- Diagnostic is higher than Descriptive.
- Predictive is higher again.
- Prescriptive appears at the highest end of both axes.
- Thus, more advanced analytics provides more value but also more complexity.

MEMORY

Descriptive -> Diagnostic -> Predictive -> Prescriptive = value and complexity rise.

Scenario 7: Structured vs Unstructured Data

Long 8-mark pattern - identify, justify, explain

SCENARIO

A smart office stores temperature values in a database table, while also collecting employee text messages, CCTV images, video clips and speech recordings.

QUESTION

Classify the data as structured or unstructured.

Best answer: Table data is Structured; media/text are Unstructured

- Structured data follows an organized format or model.
- Spreadsheets and databases are examples of structured data.
- Unstructured data does not conform to the same fixed tabular model.
- Text, images, video and speech are shown as unstructured examples.

MEMORY

Structured = Table | Unstructured = Media.

Scenario 8: Scaling Problem

Long 8-mark pattern - identify, justify, explain

SCENARIO

A smart-city project has thousands of devices continuously sending readings. The relational database grows extremely quickly and query performance becomes poor.

QUESTION

Which IoT data analytics challenge is this?

Best answer: Scaling Problem

- Large numbers of smart objects continually generate data.
- Relational databases can grow very large very quickly.
- Performance problems can appear.
- Fixing the problem may require more hardware.
- Architecture changes may also be needed and can be costly.

MEMORY

More devices -> more data -> bigger DB -> Scaling.

Scenario 9: Changing Sensor Schema

Long 8-mark pattern - identify, justify, explain

SCENARIO

An IoT application initially stores temperature and humidity. New vibration, GPS and battery fields are added frequently, so the database schema must keep changing.

QUESTION

Which challenge does this represent?

Best answer: Volatility of Data

- IoT data models may evolve over time.
- Traditional relational databases depend on a carefully designed schema.
- Frequent schema changes can affect database operation.
- IoT may require a dynamic schema.
- The slide notes that changes may even be needed daily or hourly.

MEMORY

Volatile = Variable data model.

Scenario 10: Scaling vs Volatility

Long 8-mark pattern - identify, justify, explain

SCENARIO

System A has rapidly increasing data size because millions of devices send readings. System B has manageable data size but must add new sensor fields almost every day.

QUESTION

Identify the challenge in each system.

Best answer: A = Scaling, B = Volatility

- System A is a quantity/performance growth problem -> Scaling.
- System B is a changing-data-model/schema problem -> Volatility.
- Scaling mainly concerns the growth of data and resulting database performance.
- Volatility mainly concerns frequent changes to the data model.

MEMORY

Quantity problem = Scaling | Schema-change problem = Volatility.

Scenario 11: Sensor-Side ML

Long 8-mark pattern - identify, justify, explain

SCENARIO

A vibration sensor collects data and runs the processing algorithm inside the sensor itself instead of sending all raw data to a central cloud.

QUESTION

Which learning mode is this?

Best answer: Local Learning

- Local learning collects and processes data locally.
- Processing can happen inside the sensor.
- The sensor is identified as the edge node in the slide.
- This keeps computation near the data source.

MEMORY

Local = Here | Sensor = Edge.

Scenario 12: Gateway ML

Long 8-mark pattern - identify, justify, explain

SCENARIO

Hundreds of sensors send data to a nearby IoT gateway. The gateway performs the machine-learning processing.

QUESTION

Is this Local or Remote Learning?

Best answer: Local Learning at the Fog Node

- The W12 slide treats gateway processing as local learning.
- The gateway is identified as the fog node.
- Data is processed near the source rather than in a distant centralized cloud.

MEMORY

Gateway = Fog = Local.

Scenario 13: Cloud-Based ML

Long 8-mark pattern - identify, justify, explain

SCENARIO

An IoT device collects data and sends it to a centralized data center or cloud. The ML model runs there.

QUESTION

Which learning mode is this?

Best answer: Remote Learning

- Remote learning sends collected data to a central computing unit.
- Processing may occur in a data center.
- Processing may also occur in the cloud.
- The computation is therefore away from the endpoint.

MEMORY

Remote = There | Data Center/Cloud.

Scenario 14: Local vs Remote Selection

Long 8-mark pattern - identify, justify, explain

SCENARIO

System A processes inside a sensor, System B processes at a local gateway, and System C sends data to the cloud for processing.

QUESTION

Classify all three systems.

Best answer: A = Local/Edge, B = Local/Fog, C = Remote/Cloud

- Sensor-side processing -> Local Learning at edge node.
- Gateway-side processing -> Local Learning at fog node.
- Cloud/data-center processing -> Remote Learning.
- The main distinction is whether processing stays near the IoT endpoint or moves to a central location.

MEMORY

Sensor=Edge=Local | Gateway=Fog=Local | Cloud=Remote.

Scenario 15: Mine Monitoring

Long 8-mark pattern - identify, justify, explain

SCENARIO

A mine uses sensors to continuously observe air temperature, humidity and carbon-dioxide levels and detect abnormal conditions.

QUESTION

Which major ML application domain is this?

Best answer: Monitoring

- Monitoring uses smart objects to observe the operating environment.
- The slide examples include temperature, humidity and carbon dioxide.
- ML can help identify early failure or abnormal/out-of-range behavior.
- Monitoring is the first step before control or optimization.

MEMORY

Monitoring = Observe.

Scenario 16: Automatic Ventilation

Long 8-mark pattern - identify, justify, explain

SCENARIO

The mine system detects that air-quality parameters cross a threshold and automatically increases the fresh-air flow.

QUESTION

Which ML application domain is shown?

Best answer: Behavior Control

- Behavior control commonly follows monitoring.
- A target threshold may trigger an alarm.
- More advanced systems can trigger corrective action automatically.
- Increasing fresh-air flow is one of the examples used in the slide.

MEMORY

Monitor sees -> Control acts.

Scenario 17: Purification Process Improvement

Long 8-mark pattern - identify, justify, explain

SCENARIO

A smart water plant analyzes chemical choice, temperature, stirring speed and stirring depth to reduce chemical use while maintaining purification efficiency.

QUESTION

Which ML application domain fits this scenario?

Best answer: Operations Optimization

- Operations optimization aims to improve the overall process.
- It goes beyond simple threshold-based correction.
- Analytics can combine multiple operating parameters.
- The goal is improved efficiency and improved operational results.

MEMORY

Optimization = Do it better.

Scenario 18: Self-Learning Machine

Long 8-mark pattern - identify, justify, explain

SCENARIO

A machine continuously monitors results, tests new parameter combinations, automatically applies useful optimizations and detects possible defects early.

QUESTION

Which ML application domain is this?

Best answer: Self-Healing / Self-Optimizing

- The system dynamically monitors and combines new parameters.
- It can deduce and implement new optimizations.
- It becomes self-learning and self-optimizing.
- It can detect new deviations and potential defects.
- This supports self-healing behavior.

MEMORY

Learn -> Improve -> Detect -> Heal.

Scenario 19: Full MBOS Chain

Long 8-mark pattern - identify, justify, explain

SCENARIO

A machine measures pressure, automatically reduces it when a threshold is crossed, finds better operating settings, and later learns new settings while detecting future defects.

QUESTION

Match the four tasks to the W12 ML application domains.

Best answer: Monitoring -> Behavior Control -> Operations Optimization -> Self-Healing

- Measure pressure -> Monitoring.
- Threshold corrective action -> Behavior Control.
- Find better settings -> Operations Optimization.
- Learn new optimizations and detect defects -> Self-Healing/Self-Optimizing.

MEMORY

MBOS = Monitoring - Behavior Control - Operations Optimization - Self-Healing.

Scenario 20: High-Speed Sensor Stream

Long 8-mark pattern - identify, justify, explain

SCENARIO

Thousands of sensors send readings every millisecond, and the analytics system must collect and analyze them rapidly.

QUESTION

Which Big Data V is most relevant?

Best answer: Velocity

- Velocity refers to how quickly data is collected and analyzed.
- It focuses on data-arrival and processing speed.
- Very rapid IoT streams are a classic velocity issue.

MEMORY

Velocity = Speed.

Scenario 21: Many Data Formats

Long 8-mark pattern - identify, justify, explain

SCENARIO

An IoT platform receives database records, semi-structured messages, video, images and text.

QUESTION

Which Big Data V is most relevant?

Best answer: Variety

- Variety refers to different types of data.
- The slide mentions structured, semi-structured and unstructured data.
- Mixed IoT formats therefore represent a variety challenge.

MEMORY

Variety = Type.

Scenario 22: Petabytes of IoT History

Long 8-mark pattern - identify, justify, explain

SCENARIO

A smart-city system stores several petabytes of historical IoT readings.

QUESTION

Which Big Data V does this demonstrate?

Best answer: Volume

- Volume refers to the scale or total amount of data.
- The slide gives a range from gigabytes to petabytes and even exabytes.
- Petabyte-scale historical storage is therefore a volume issue.

MEMORY

Volume = Size.

Scenario 23: Full 3V Scenario

Long 8-mark pattern - identify, justify, explain

SCENARIO

A city receives IoT data at extremely high speed, in many different formats, and in extremely large total quantities.

QUESTION

Map the three clues to the 3V model.

Best answer: Speed=Velocity, Formats=Variety, Size=Volume

- High arrival/processing speed -> Velocity.
- Different data forms -> Variety.
- Large total scale -> Volume.
- The 3V memory is Speed - Type - Size.

MEMORY

Velocity=Speed | Variety=Type | Volume=Size.

Security Scenario Practice

Identify the listed W12 IoT security challenge

1. Weak CPU/Memory Device

Advanced protection is difficult because the device has very low computation power.

Answer: Low Processing Power

2. Old Industrial Equipment

A connected factory device was designed years ago and lacks modern security features.

Answer: Legacy Assets

3. No Firmware Patches

An IoT camera has not received firmware updates for a long time.

Answer: Firmware Updates Missing

4. Plaintext Sensitive Data

A smart sensor transmits sensitive information without encryption.

Answer: Lack of Encryption

5. Compromised Device Army

Thousands of infected IoT devices are controlled together for malicious activity.

Answer: Botnet Attacks

6. Fake Sensor

An unauthorized counterfeit device enters the IoT network and pretends to be legitimate.

Answer: Rogue/Counterfeit IoT Devices

7. Unsafe User Practice

Users configure devices insecurely because they do not understand security practices.

Answer: Lack of Awareness in Users

8. Different Vendor Rules

Devices from multiple vendors follow inconsistent security rules and protections.

Answer: Inconsistent Security Standards

SECURITY MEMORY

Standards + Low Power + Legacy + Awareness + Botnet + Encryption + Firmware + Rogue Devices

Hard Integrated Scenario 1

Smart Factory - almost the whole analytics chapter

SCENARIO

A smart factory must show current status, find fault causes, predict future failures, recommend actions, process some data at a gateway, optimize operations, self-detect defects, and handle huge/fast/varied data.

Current machine status	->	**Descriptive**
Why fault happened	->	**Diagnostic**
Future failure estimate	->	**Predictive**
Recommended action	->	**Prescriptive**
Gateway processing	->	**Local Learning / Fog**
Improve efficiency	->	**Operations Optimization**
Automatic learning/defect detection	->	**Self-Healing / Self-Optimizing**

FULL-CHAPTER MEMORY

DDPP + Local/Fog + MBOS + 3V

Fast arrival	->	**Velocity**
Many formats	->	**Variety**

Hard Integrated Scenario 2

Smart Mine - analytics + learning + Big Data

SCENARIO

A mine measures temperature, humidity and CO2. It generates an alarm and increases ventilation at a threshold, sends selected data to the cloud for learning, and stores very large quantities of different sensor and media data.

Observe environment	->	**Monitoring**
Threshold alarm/action	->	**Behavior Control**
Cloud-based learning	->	**Remote Learning**
Large amount of data	->	**Volume**

FAST IDENTIFY

Observe=Monitoring | Act=Behavior Control | Cloud=Remote | Size=Volume | Formats=Variety

Hard Trap Questions

True/False + correction

1. Descriptive Analytics answers why something happened. **FALSE - Diagnostic answers 'Why?'**

2. Predictive Analytics recommends the best action. **FALSE - Prescriptive recommends action.**

3. Prescriptive is shown as more complex than Descriptive. **TRUE.**

4. Images and video are structured data. **FALSE - they are unstructured.**

5. Scaling means the schema keeps changing. **FALSE - that is Volatility.**

6. Local Learning can run at an edge sensor. **TRUE.**

7. Local Learning can also run at a fog gateway. **TRUE.**

8. Remote Learning processes data in the cloud/data center. **TRUE.**

9. Behavior Control may trigger corrective action. **TRUE.**

10. Velocity means different types of data. **FALSE - Variety means different types.**

11. Volume means data scale. **TRUE.**

12. Missing firmware updates are a security challenge. **TRUE.**

Rapid Practice - 30 Questions

Try without looking

1. Name four analytics types.
2. Descriptive answers what?
3. Diagnostic answers what?
4. Predictive answers what?
5. Prescriptive answers what?
6. Which analytics type is highest in value/complexity?
7. Give two structured-data examples.
8. Give four unstructured-data examples.
9. What is Scaling Problem?
10. What is Data Volatility?
11. Why might dynamic schema be needed?
12. Name ML learning categories shown.
13. What is Local Learning?
14. Where can Local Learning run?
15. What is Remote Learning?
16. Where does Remote Learning run?
17. Name four ML application domains.
18. What is Monitoring?
19. What is Behavior Control?
20. What is Operations Optimization?
21. What is Self-Healing?
22. Name Big Data 3V.
23. What is Velocity?
24. What is Variety?
25. What is Volume?
26. Name four Big Data tools.
27. What is Edge Streaming Analytics?
28. Name four IoT security challenges.
29. What is a Botnet Attack?
30. Why are firmware updates important?

Final Exam Strategy

How to solve an unseen W12 scenario quickly

MASTER DECISION MAP

Current condition	->	Descriptive	Data Center/Cloud	->	Remote Learning
Find reason	->	Diagnostic	Observe	->	Monitoring
Future outcome	->	Predictive	Threshold -> action	->	Behavior Control
Recommended action	->	Prescriptive	Improve process	->	Operations Optimization
DB grows too fast	->	Scaling	Auto learn/fix	->	Self-Healing
Schema changes often	->	Volatility	Speed / Type / Size	->	Velocity / Variety / Volume
Sensor/Gateway	->	Local Learning			

8-MARK ANSWER FORMULA

- 1. Identify the exact analytics/security concept.
- 2. Write a one-line definition.
- 3. Link the scenario clue to the concept.
- 4. Write 4-7 clean slide-based points.
- 5. Add one comparison/example where relevant.
- 6. Finish with a short conclusion.

FINAL MEMORY

DDPP -> SV -> Local/Remote -> MBOS -> 3V -> Security 8